

CHE 435/525 *Process Systems Analysis and Control*

Lecture 7: **First-Order Systems and Integrating Systems**

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January 30, 2025



Recapture: Laplace transform and transfer function

- Input-output model: Transfer function

$$G(s) = Y(s)/U(s)$$

- Calculating output response:

$$u(t) \xrightarrow{\text{L.T.}} U(s) \xrightarrow{\text{T.F.}} Y(s) \xrightarrow{\text{P.F.E.}} y(t)$$

- **Qualitative analysis** possible! Examine the roots on the denominator
 - real part (growth/decay) and imaginary part (oscillations)

Learning objectives of this lecture

- ① Explain what kind of physical processes are **first-order systems** and **integrating systems**
- ② Calculate the response of first-order systems to **standard inputs**
- ③ Indicate the characteristic quantities on the response curve and relate them to the model parameters

Example 1: Liquid storage tank

- Model

$$A \frac{dh}{dt} = q_i - \frac{1}{R} h$$

- Discuss: Interpret this model
- Transfer function from $u = q_i$ to $y = h$?

$$G(s) = \frac{1}{As + 1/R} \quad \left(= \frac{R}{ARs + 1} \right)$$

Example 2: Tank heated with steam

- Model

$$\rho VC_P \frac{dT}{dt} = UA(T_s - T)$$

- Discuss: Interpret this model
- Transfer function from $u = T_s$ to $y = T$?

$$G(s) = \frac{1}{\frac{\rho VC_P}{UA}s + 1}$$

- Exercise: Develop a model for a thermometer.
 - ▶ Formally the same. Just different interpretations for these symbols.

First-order system

- Physical picture:
 - ▶ A single balance equation: $\text{LHS} = dy/dt$, RHS contains input.
 - ▶ The input **directly** affects the evolution of the output.
 - ▶ The system consists of “**resistance and capacitance**”.
- General form of transfer function

$$G(s) = \frac{K}{\tau s + 1}.$$

- ▶ τ is the **time constant**
- ▶ $K = G(0)$ is called the **static gain**

Analyzing T.F. with standard inputs

- Standard inputs:

- ▶ Step input $u(t) = \begin{cases} 0, & t < 0 \\ M, & t \geq 0 \end{cases}$

- ▶ Impulse input $u(t) = M\delta(t)$

- ▶ Ramp input $u(t) = \begin{cases} 0, & t < 0 \\ Mt, & t \geq 0 \end{cases}$

- ▶ Sinusoidal input $u(t) = \begin{cases} 0, & t < 0 \\ A \sin \omega t, & t \geq 0 \end{cases}$

- Previously, we introduced the following routine of analysis

$$u(t) \xrightarrow{\text{L.T.}} U(s) \xrightarrow{\text{T.F.}} Y(s) \xrightarrow{\text{P.F.E.}} y(t)$$

Calculating step response

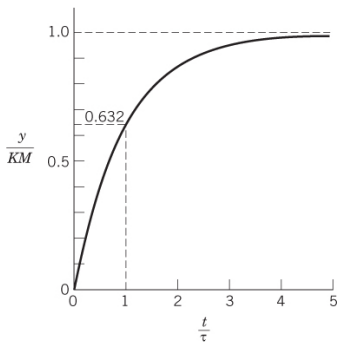
- Step input $u(t) = M$ ($t \geq 0$). By L.T., $U(s) = M/s$.
- By T.F. model $G(s) = K/(\tau s + 1)$, output:

$$Y(s) = G(s)U(s) = \frac{KM}{s(\tau s + 1)} = \frac{KM}{\tau} \frac{1}{s(s + \frac{1}{\tau})}$$

- By P.F.E.,

$$Y(s) = KM \left(\frac{1}{s} - \frac{1}{s + (1/\tau)} \right) \Rightarrow y(t) = KM(1 - e^{-t/\tau})$$

Characteristics of step response



t	$y(t)/KM = 1 - e^{-t/\tau}$
0	0
1τ	0.6321
2τ	0.8647
3τ	0.9502
4τ	0.9817
5τ	0.9933

- ① At $t = 0$, the slope suddenly changes to KM/τ
- ② The curve is **monotonic**
- ③ As $t \rightarrow \infty$, the limit is KM . **Static gain = K**

Calculating impulse response

- Step input $u(t) = M\delta(t)$ ($t \geq 0$). By L.T., $U(s) = M$.
- By T.F. model $G(s) = K/(\tau s + 1)$, output:

$$Y(s) = G(s)U(s) = \frac{KM}{\tau s + 1} \xrightarrow{\text{P.F.E.}} y(t) = \frac{KM}{\tau} e^{-t/\tau}$$

- Plot this curve and describe its appearance

Observations from step and impulse responses

- When $M = 1$, $U(s) = \mathcal{L}[\delta(t)] = 1$ (unit impulse input), we simply get $Y(s) = G(s)$.
 - ▶ A transfer function is the L.T. of its unit impulse response.

$$y_{\text{unit-impulse}}(t) = g(t) = \mathcal{L}^{-1}[G(s)]$$

- ▶ This is a general conclusion for all linear systems (not only applicable to first-order systems).
- $u_{\text{impulse}} = du_{\text{step}}/dt$, $y_{\text{impulse}} = dy_{\text{step}}/dt$. Is this a coincidence?
 - ▶ No! Since $U_{\text{impulse}}(s) = sU_{\text{step}}(s)$, multiplying by $G(s)$, we get $Y_{\text{impulse}}(s) = sY_{\text{step}}(s)$.
- How to calculate ramp response? ($y_{\text{ramp}} = Mt$ for $t \geq 0$)

** Calculating response to ANY input signal

Convolution formula

The product of two L.T.s is the L.T. of the **convoluted signal**:

$$\mathcal{L}[f_1(t)] \cdot \mathcal{L}[f_2(t)] = \mathcal{L} \left[\int_0^t f_1(t - \tau) f_2(\tau) d\tau \right]$$

- Given the transfer function $G(s)$ and input signal $u(t)$, the output is $Y(s) = G(s)U(s)$
- By convolution formula, $G(s)U(s)$ is the L.T. of the convolution of $g(t)$ and $u(t)$.
 - ▶ $g(t)$: inverse L.T. of $G(s)$, exactly the **unit impulse response**
- Thus,

$$y(t) = \int_0^t g(t - \tau) u(\tau) d\tau.$$

Integrating system

- Consider the tank without exit (resistance $R \rightarrow \infty$):

$$A \frac{dh}{dt} = q_i$$

- T.F. from $u = q_i$ to $y = h$ is

$$G(s) = K/s \quad (\text{where } K = 1/A)$$

- The relation between the input and output is purely an **integrator**

$$y(t) = K \int_0^t u(\tau) d\tau$$

- Step response?
- Impulse response?